Subject: ASD&D **Semester:** V

Class: TE **Batch:** T11

Name: Ananya Amesur **Roll No:** 2401005

Experiment 9

Aim: To provision cloud infrastructure using Terraform.

Lab Outcome: LO4

LO4: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

Theory:

Infrastructure as Code (IaC)

Infrastructure as Code (IaC) is the practice of managing, provisioning, and configuring compute, storage, and networking resources through machine-readable definition files rather than physical hardware configuration or interactive configuration tools. IaC ensures reproducibility, consistency, rapid scaling, and version control across multiple environments, eliminating configuration drift and manual operational errors.

HashiCorp Terraform Overview

Terraform is an open-source, cloud-agnostic IaC software tool created by HashiCorp. It enables users to define both cloud and on-premise resources using a declarative configuration language known as HashiCorp Configuration Language (HCL).

- Declarative Approach: Rather than writing procedural scripts describing *how* to set up infrastructure step by step, developers declare the *desired end-state* of the target environment. Terraform determines the dependency graph and executes the actions required to achieve that state.
- State Management: Terraform maintains a state file (terraform.tfstate) that maps real-world provisioned cloud resources to the local configuration definitions. It uses this state to calculate changes, manage drift, and safely destroy resources.

Core Terraform Workflow & Commands

- terraform init: Initializes the working directory containing Terraform configuration files. It downloads the required provider plugins (such as hashicorp/aws) and sets up the local state backend.
- terraform validate: Verifies the syntax and internal consistency of the configuration files within the directory without accessing remote APIs.
- terraform plan: Generates an execution plan by comparing the declared configuration files with the current state file and remote cloud resources. It previews the exact resources to be created, modified, or destroyed without executing changes.
- terraform apply: Executes the proposed changes outlined in the execution plan to provision the desired infrastructure on the cloud provider.
- terraform output: Extracts and displays structured values defined in outputs.tf (such as assigned public IP addresses and instance IDs) directly from the state file.
- terraform destroy: Reads the current state file and safely deprovisions all managed infrastructure in the reverse order of dependency, preventing unnecessary cloud resource costs.

AWS Provider & Dynamic Data Sources

- AWS Provider: Authenticates with the AWS cloud platform via API credentials (Access Key ID and Secret Access Key) and targets a designated region (e.g., ap-southeast-2).
- Data Sources (data blocks): Enable dynamic querying of existing external resources. Instead of hardcoding static Amazon Machine Image (AMI) IDs or network subnet identifiers, Terraform dynamically fetches them using AWS Systems Manager (SSM) Parameter Store and default VPC data queries, making the code portable and resilient to AMI deprecation.

Declarative Configuration Code:

The configuration code was organized inside C:\Terraform.

1. main.tf (Infrastructure Definition)

Teraform

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 6.0"
    }
  }
}

provider "aws" {
  region = "ap-southeast-2"
}

data "aws_ssm_parameter" "ubuntu_ami" {
  name = "/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id"
}

data "aws_vpc" "default" {
  default = true
}

data "aws_subnets" "default" {
  filter {
    name   = "vpc-id"
    values = [data.aws_vpc.default.id]
  }
}

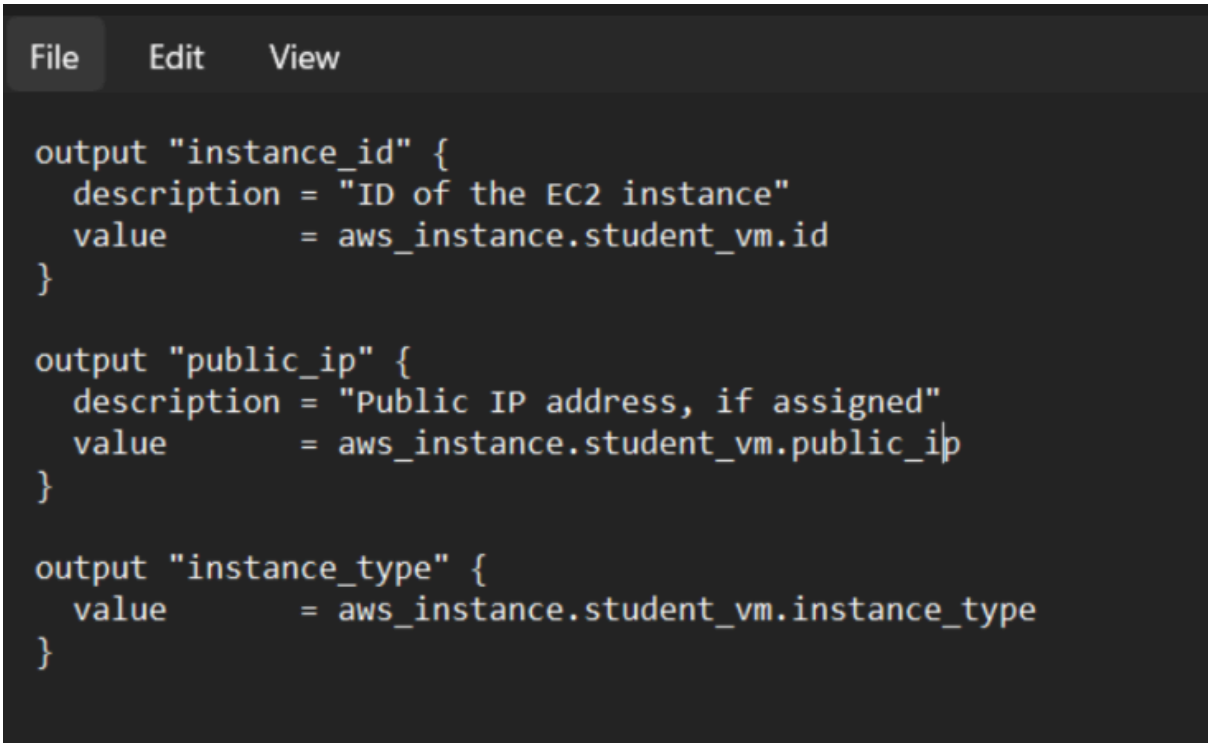
resource "aws_instance" "student_vm" {
  ami           = data.aws_ssm_parameter.ubuntu_ami.value
  instance_type = "t3.micro"
  subnet_id     = data.aws_subnets.default.ids[0]
  key_name      = "vmb-key"

  tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
```

```
  volume_tags = {
    Name        = "Terraform-Student-VM"
    Environment = "Dev"
    Project     = "Experiment7"
  }
}
```

2. outputs.tf (Output State Extraction)

Terraform



Implementation:

Phase 1: Local Environment & Credentials Configuration

Step 1: AWS IAM Credentials & Access Key Setup

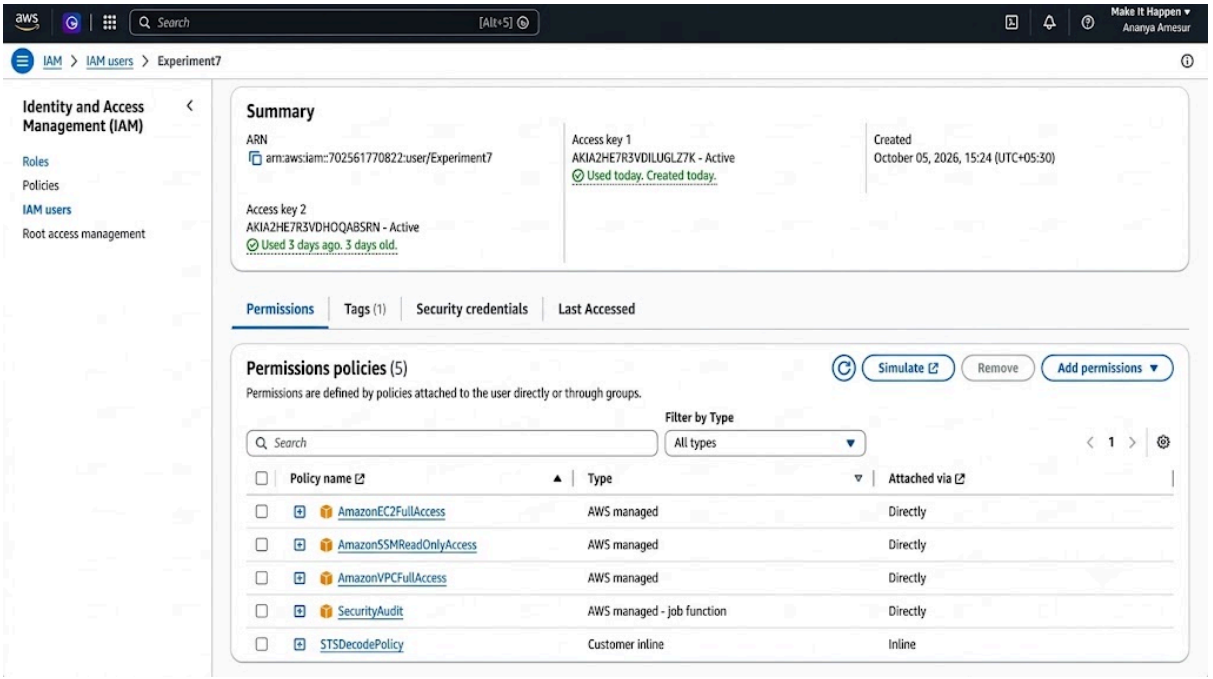
1. Log in to the AWS Management Console and navigate to IAM > Users > Experiment7.

2. Select Security credentials, click Create access key, and choose Command Line Interface (CLI).

3. Generate and record the credentials:

- **Access Key ID:** AKIA2HE7R3VDILU6LZ7K

- **Secret Access Key:**
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



Step 2: AWS CLI Authentication & Verification

1. Open Windows PowerShell and run aws configure:

PowerShell

AWS Access Key ID: AKIA2HE7R3VDILU6LZ7K

AWS Secret Access Key: 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

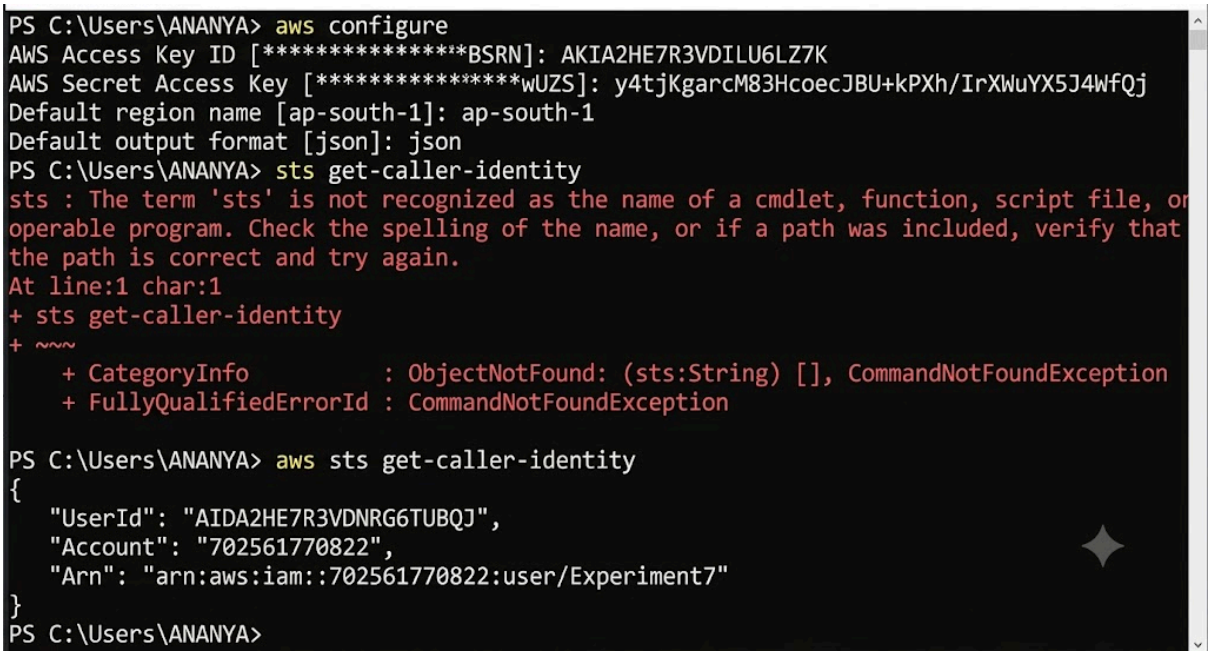
Default region name: ap-southeast-2

Default output format: json

2. Verify identity via STS:

PowerShell

aws sts get-caller-identity



.

Step 3: EC2 Key Pair Generation in ap-southeast-2

1. Navigate to AWS Console > EC2 > Key Pairs in region ap-southeast-2.

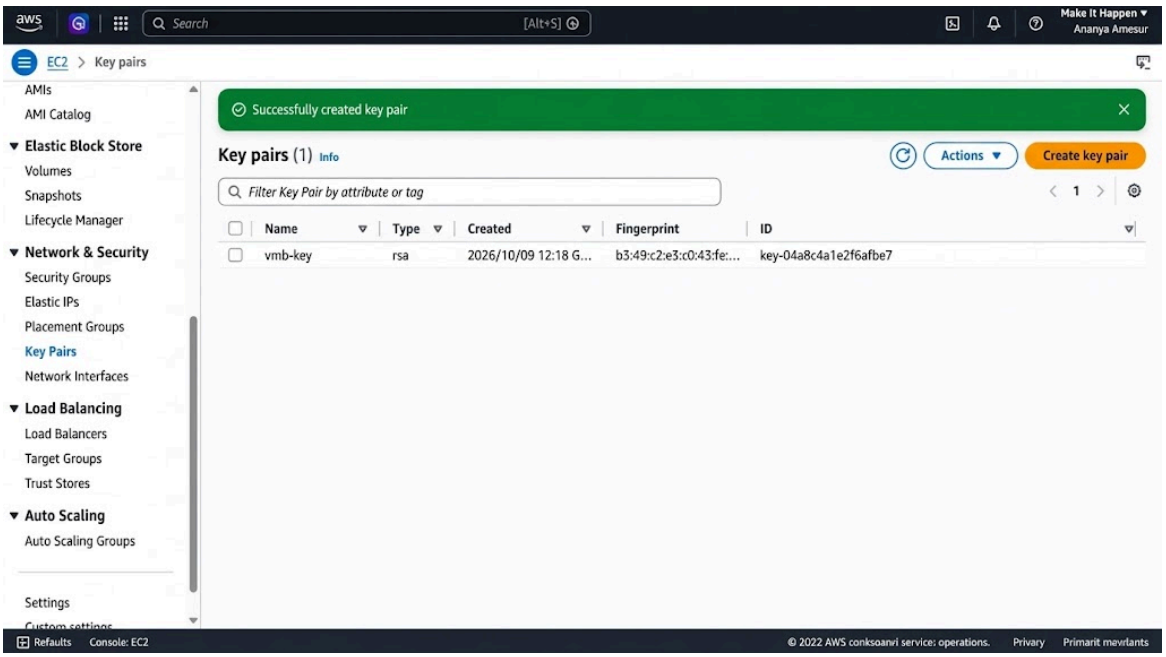
2. Click Create key pair:

    o Name: vmb-key

    o Key pair type: RSA

    o Private key file format: .pem

3. Download and save vmb-key.pem.



Phase 2: Resolving IAM Permissions & Policy Extensions

Step 4: Attaching SSM Read-Only Permissions

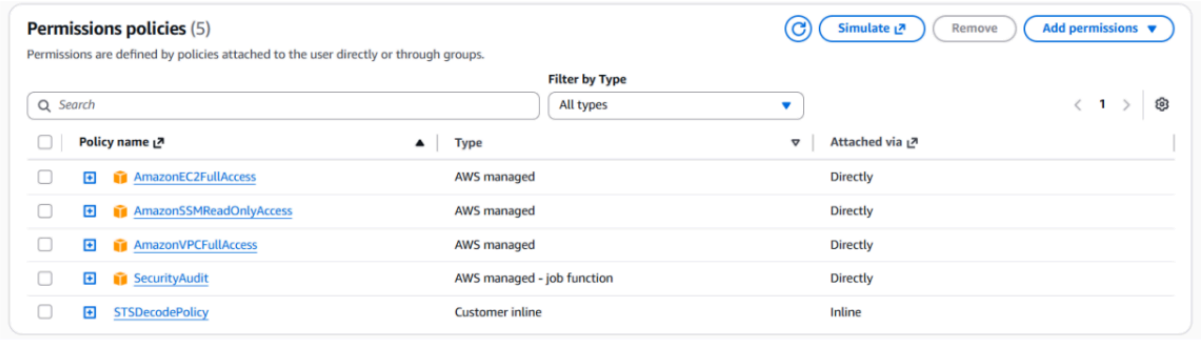
To enable Terraform to query the Canonical Ubuntu AMI ID dynamically from AWS Parameter Store, attach the SSM policy:

1. Go to IAM > Users > Experiment7 > Permissions.

2. Click Add permissions > Attach policies directly.

3. Search and select AmazonSSMReadOnlyAccess.

4. Click Add permissions.



Phase 3: Terraform Execution Lifecycle

Step 5: Directory Setup & Terraform Initialization

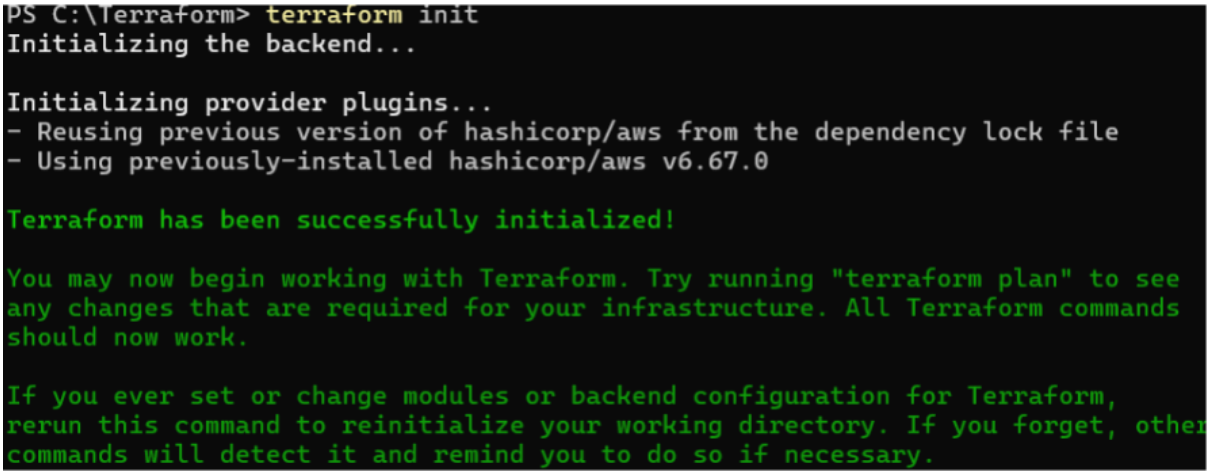
1. Create directory C:\Terraform and save main.tf and outputs.tf inside.

2. Open PowerShell, navigate to directory, and initialize backend:

PowerShell

cd C:\Terraform

terraform init



Step 6: Code Validation & Execution Planning

1. Validate syntax:

PowerShell

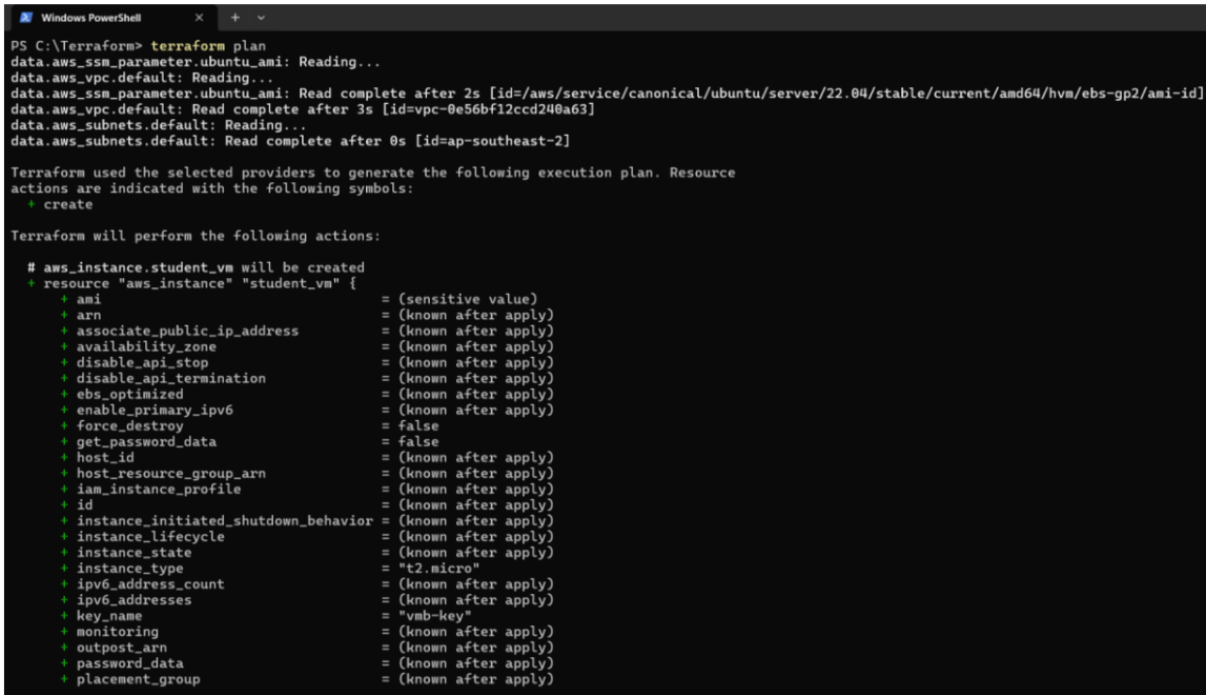
terraform validate

2. Generate execution plan:

PowerShell

terraform plan

```
PS C:\Terraform> terraform validate
Success! The configuration is valid.
```



```
Windows PowerShell
PS C:\Terraform> terraform plan
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_vpc.default: Reading...
data.aws_ssm_parameter.ubuntu_ami: Read complete after 2s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
data.aws_vpc.default: Read complete after 3s [id=vpc-0e56bf12ccd240a63]
data.aws_subnets.default: Reading...
data.aws_subnets.default: Read complete after 0s [id=ap-southeast-2]

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.student_vm will be created
  + resource "aws_instance" "student_vm" {
      + ami                                  = (sensitive value)
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
      + host_resource_group_arn              = (known after apply)
      + iam_instance_profile                 = (known after apply)
      + id                                   = (known after apply)
      + instance_initiated_shutdown_behavior = (known after apply)
      + instance_lifecycle                   = (known after apply)
      + instance_state                       = (known after apply)
      + instance_type                        = "t2.micro"
      + ipv6_address_count                   = (known after apply)
      + ipv6_addresses                       = (known after apply)
      + key_name                             = "vmb-key"
      + monitoring                           = (known after apply)
      + outpost_arn                          = (known after apply)
      + password_data                        = (known after apply)
      + placement_group                      = (known after apply)
```

```
      + placement_group_id                 = (known after apply)
      + placement_partition_number         = (known after apply)
      + primary_network_interface_id       = (known after apply)
      + private_dns                        = (known after apply)
      + private_ip                         = (known after apply)
      + public_dns                         = (known after apply)
      + public_ip                          = (known after apply)
      + region                             = "ap-southeast-2"
      + secondary_private_ips              = (known after apply)
      + security_groups                    = (known after apply)
      + source_dest_check                  = true
      + spot_instance_request_id           = (known after apply)
      + subnet_id                          = "subnet-0ecc646eb242b088a"
      + tags                               = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tags_all                           = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + tenancy                            = (known after apply)
      + user_data_base64                   = (known after apply)
      + user_data_replace_on_change        = false
      + volume_tags                        = {
          + "Environment" = "Dev"
          + "Name"        = "Terraform-Student-VM"
          + "Project"     = "Experiment7"
        }
      + vpc_security_group_ids             = (known after apply)

      + capacity_reservation_specification (known after apply)

      + cpu_options (known after apply)

      + ebs_block_device (known after apply)
```

```
      + enclave_options (known after apply)

      + ephemeral_block_device (known after apply)

      + instance_market_options (known after apply)

      + maintenance_options (known after apply)

      + metadata_options (known after apply)

      + network_interface (known after apply)

      + primary_network_interface (known after apply)

      + private_dns_name_options (known after apply)

      + root_block_device (known after apply)

      + secondary_network_interface (known after apply)
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + instance_id   = (known after apply)
  + public_ip     = (known after apply)

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Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to
take exactly these actions if you run "terraform apply" now.
```

Step 7: Applying Infrastructure Provisioning

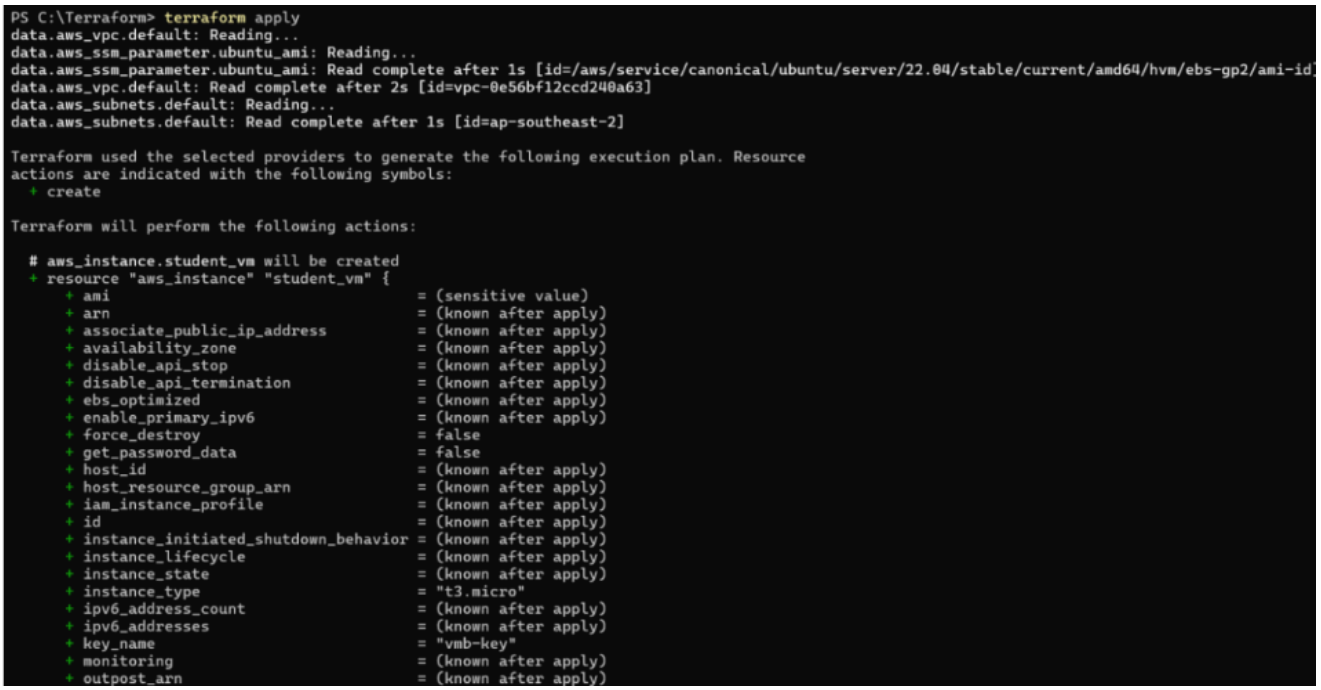
1. Execute apply command:

PowerShell

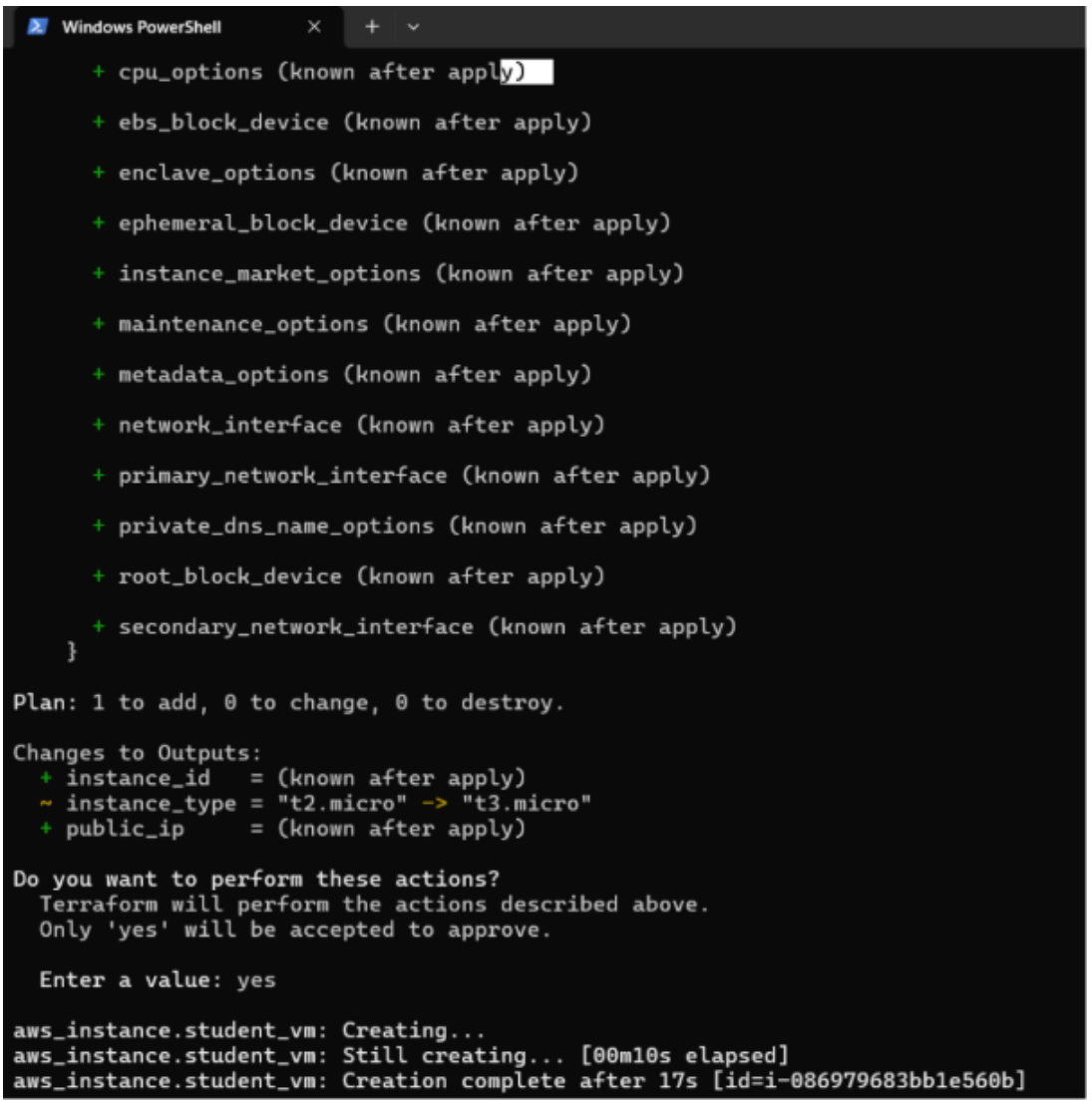
terraform apply

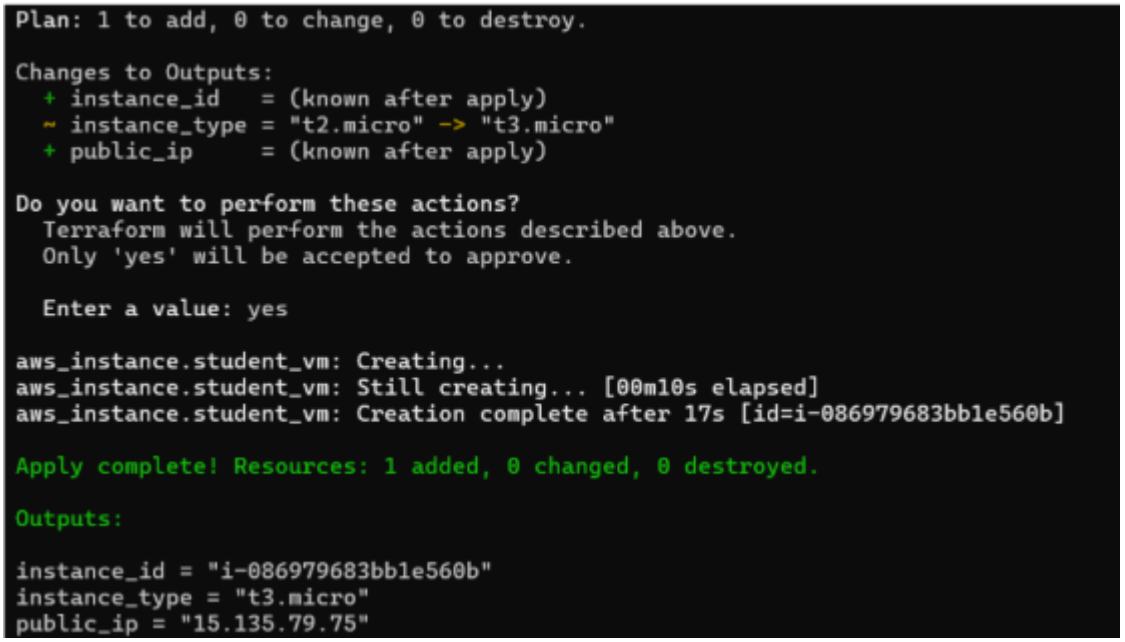
2. Enter yes at the approval prompt.

3. Confirm resource creation completion.




Windows PowerShell
+ monitoring = (known after apply)
+ outpost_arn = (known after apply)
+ password_data = (known after apply)
+ placement_group = (known after apply)
+ placement_group_id = (known after apply)
+ placement_partition_number = (known after apply)
+ primary_network_interface_id = (known after apply)
+ private_dns = (known after apply)
+ private_ip = (known after apply)
+ public_dns = (known after apply)
+ public_ip = (known after apply)
+ region = "ap-southeast-2"
+ secondary_private_ips = (known after apply)
+ security_groups = (known after apply)
+ source_dest_check = true
+ spot_instance_request_id = (known after apply)
+ subnet_id = "subnet-0ecc646eb242b088a"
+ tags = {
+ "Environment" = "Dev"
+ "Name" = "Terraform-Student-VM"
+ "Project" = "Experiment7"
}
+ tags_all = {
+ "Environment" = "Dev"
+ "Name" = "Terraform-Student-VM"
+ "Project" = "Experiment7"
}
+ tenancy = (known after apply)
+ user_data_base64 = (known after apply)
+ user_data_replace_on_change = false
+ volume_tags = {
+ "Environment" = "Dev"
+ "Name" = "Terraform-Student-VM"
+ "Project" = "Experiment7"
}
+ vpc_security_group_ids = (known after apply)
+ capacity_reservation_specification (known after apply)
+ cpu_options (known after apply)


Windows PowerShell
+ cpu_options (known after apply)
+ ebs_block_device (known after apply)
+ enclave_options (known after apply)
+ ephemeral_block_device (known after apply)
+ instance_market_options (known after apply)
+ maintenance_options (known after apply)
+ metadata_options (known after apply)
+ network_interface (known after apply)
+ primary_network_interface (known after apply)
+ private_dns_name_options (known after apply)
+ root_block_device (known after apply)
+ secondary_network_interface (known after apply)
}
Plan: 1 to add, 0 to change, 0 to destroy.
Changes to Outputs:
+ instance_id = (known after apply)
~ instance_type = "t2.micro" -> "t3.micro"
+ public_ip = (known after apply)
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
aws_instance.student_vm: Creating...
aws_instance.student_vm: Still creating... [00m10s elapsed]
aws_instance.student_vm: Creation complete after 17s [id=i-086979683bb1e560b]


Plan: 1 to add, 0 to change, 0 to destroy.
Changes to Outputs:
+ instance_id = (known after apply)
~ instance_type = "t2.micro" -> "t3.micro"
+ public_ip = (known after apply)
Do you want to perform these actions?
Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.
Enter a value: yes
aws_instance.student_vm: Creating...
aws_instance.student_vm: Still creating... [00m10s elapsed]
aws_instance.student_vm: Creation complete after 17s [id=i-086979683bb1e560b]
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
Outputs:
instance_id = "i-086979683bb1e560b"
instance_type = "t3.micro"
public_ip = "15.135.79.75"

Phase 4: Verification & Teardown

Step 8: Verifying Terraform Outputs & AWS EC2 Console State

1. Run output command in PowerShell:

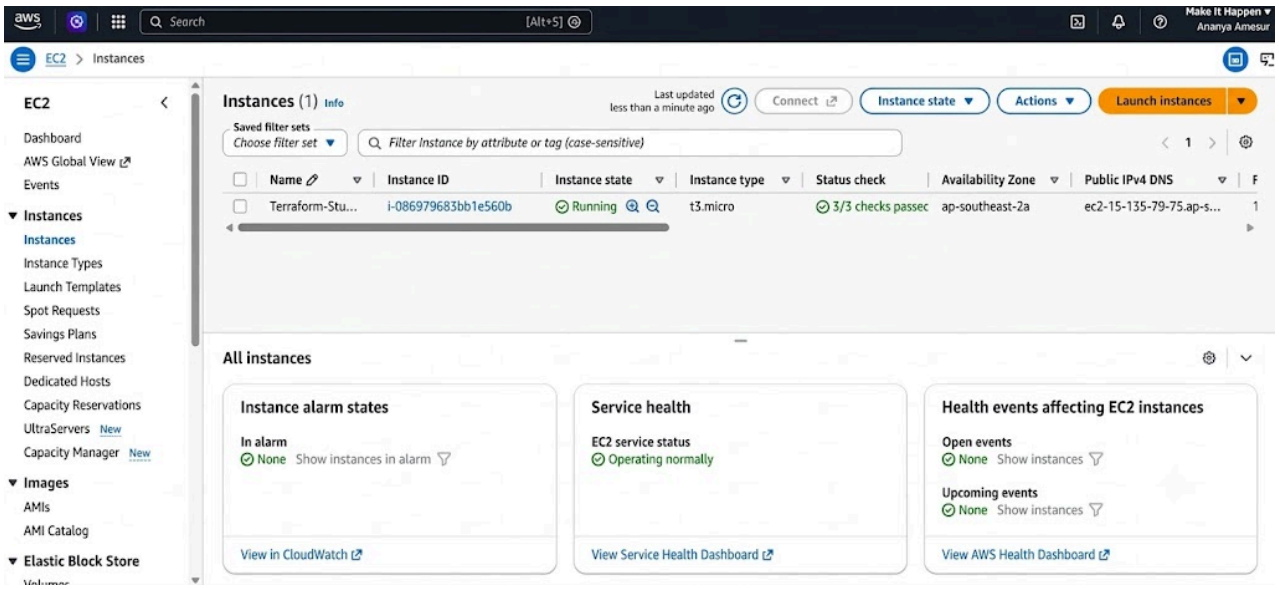
PowerShell

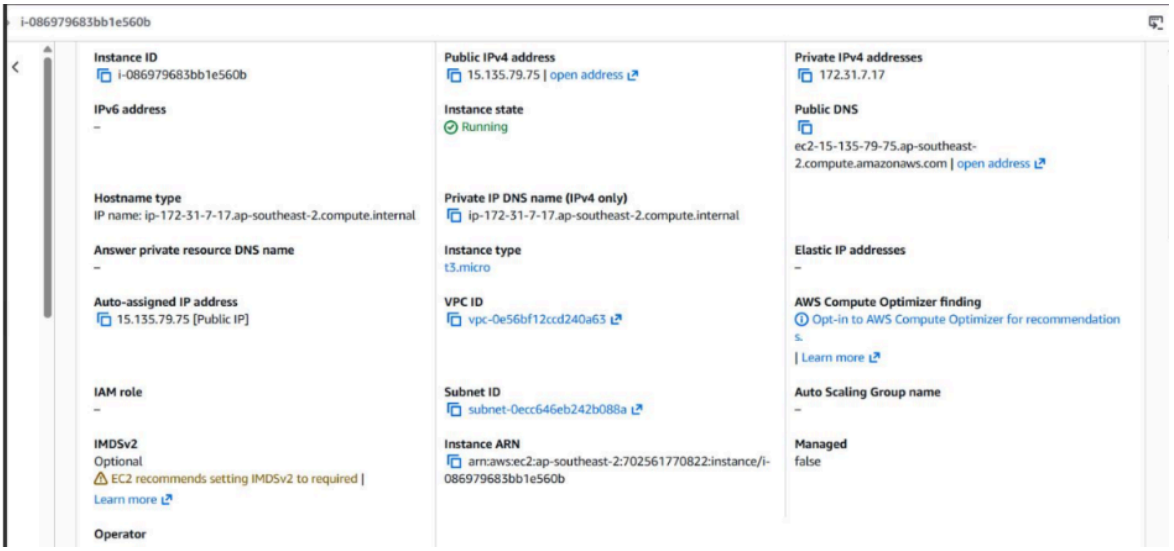
terraform output

2. Open browser and navigate to AWS EC2 Dashboard > Instances in region ap southeast-2.

3. Verify that Terraform-Student-VM (i-086979683bb1e560b) is running with public IP 15.135.79.75.

```
PS C:\Terraform> terraform output
instance_id = "i-086979683bb1e560b"
instance_type = "t3.micro"
public_ip = "15.135.79.75"
```





Step 9: Deprovisioning Cloud Resources (terraform destroy)

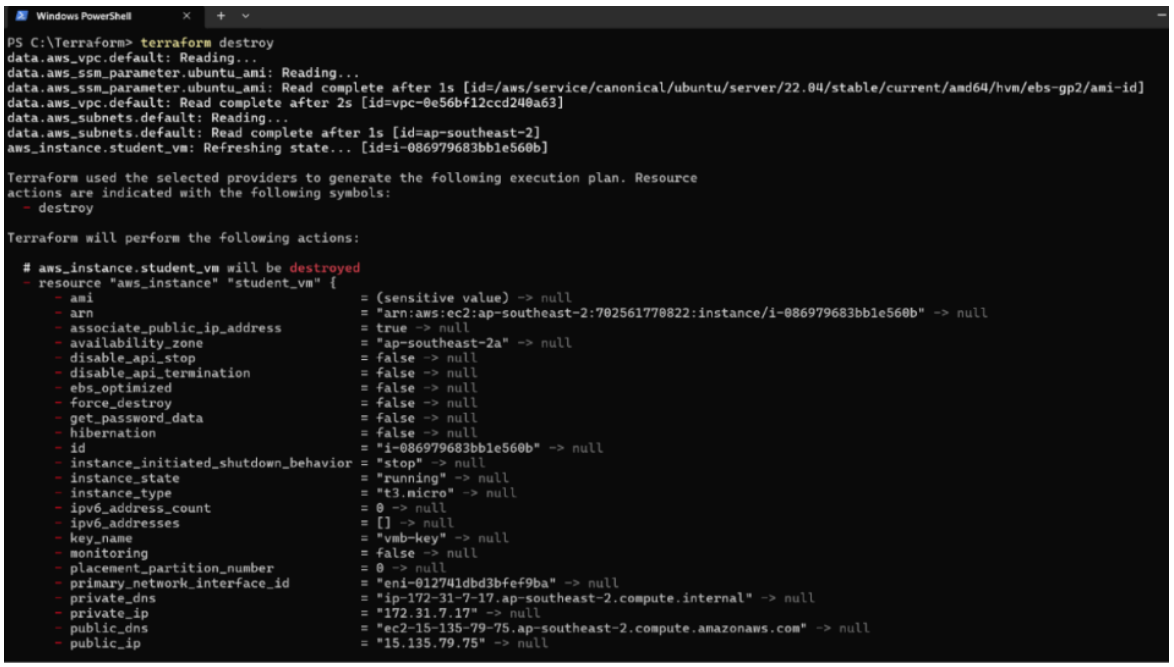
1. In PowerShell, initiate teardown:

PowerShell

terraform destroy

2. Type yes when prompted.

3. Confirm resource destruction.



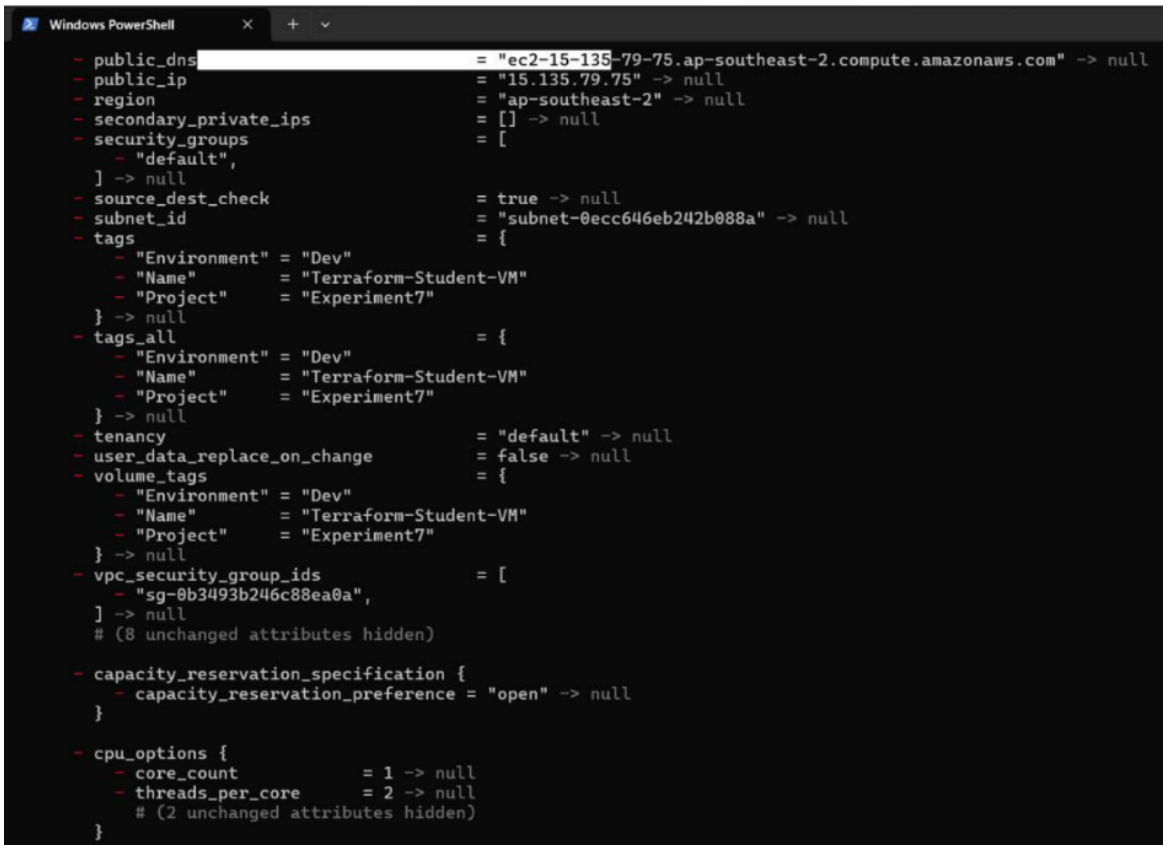
```
Windows PowerShell

PS C:\Terraform> terraform destroy
data.aws_vpc.default: Reading...
data.aws_ssm_parameter.ubuntu_ami: Reading...
data.aws_ssm_parameter.ubuntu_ami: Read complete after 1s [id=/aws/service/canonical/ubuntu/server/22.04/stable/current/amd64/hvm/ebs-gp2/ami-id]
data.aws_vpc.default: Read complete after 2s [id=vpc-0e56bf12ccd240a63]
data.aws_subnets.default: Reading...
data.aws_subnets.default: Read complete after 1s [id=ap-southeast-2]
aws_instance.student_vm: Refreshing state... [id=i-086979683bb1e560b]

Terraform used the selected providers to generate the following execution plan. Resource
actions are indicated with the following symbols:
  - destroy

Terraform will perform the following actions:

  # aws_instance.student_vm will be destroyed
  - resource "aws_instance" "student_vm" {
      - ami                                  = (sensitive value) -> null
      - arn                                  = "arn:aws:ec2:ap-southeast-2:702561770822:instance/i-086979683bb1e560b" -> null
      - associate_public_ip_address          = true -> null
      - availability_zone                    = "ap-southeast-2a" -> null
      - disable_api_stop                     = false -> null
      - disable_api_termination              = false -> null
      - ebs_optimized                        = false -> null
      - force_destroy                        = false -> null
      - get_password_data                    = false -> null
      - hibernation                          = false -> null
      - id                                   = "i-086979683bb1e560b" -> null
      - instance_initiated_shutdown_behavior = "stop" -> null
      - instance_state                       = "running" -> null
      - instance_type                        = "t3.micro" -> null
      - ipv6_address_count                   = 0 -> null
      - ipv6_addresses                       = [] -> null
      - key_name                             = "vmb-key" -> null
      - monitoring                           = false -> null
      - placement_partition_number           = 0 -> null
      - primary_network_interface_id         = "eni-012741dbd3bfef9ba" -> null
      - private_dns                          = "ip-172-31-7-17.ap-southeast-2.compute.internal" -> null
      - private_ip                           = "172.31.7.17" -> null
      - public_dns                           = "ec2-15-135-79-75.ap-southeast-2.compute.amazonaws.com" -> null
      - public_ip                            = "15.135.79.75" -> null
```



```
Windows PowerShell

      - public_dns                           = "ec2-15-135-79-75.ap-southeast-2.compute.amazonaws.com" -> null
      - public_ip                            = "15.135.79.75" -> null
      - region                               = "ap-southeast-2" -> null
      - secondary_private_ips                = [] -> null
      - security_groups                      = [
          - "default",
        ] -> null
      - source_dest_check                    = true -> null
      - subnet_id                            = "subnet-0ecc646eb242b088a" -> null
      - tags                                 = {
          - "Environment" = "Dev"
          - "Name"        = "Terraform-Student-VM"
          - "Project"     = "Experiment7"
        } -> null
      - tags_all                             = {
          - "Environment" = "Dev"
          - "Name"        = "Terraform-Student-VM"
          - "Project"     = "Experiment7"
        } -> null
      - tenancy                              = "default" -> null
      - user_data_replace_on_change          = false -> null
      - volume_tags                          = {
          - "Environment" = "Dev"
          - "Name"        = "Terraform-Student-VM"
          - "Project"     = "Experiment7"
        } -> null
      - vpc_security_group_ids               = [
          - "sg-0b3493b246c88ea0a",
        ] -> null
        # (8 unchanged attributes hidden)

      - capacity_reservation_specification {
          - capacity_reservation_preference = "open" -> null
        }

      - cpu_options {
          - core_count            = 1 -> null
          - threads_per_core      = 2 -> null
            # (2 unchanged attributes hidden)
        }
```

```
    }

  - credit_specification {
      - cpu_credits = "unlimited" -> null
    }

  - enclave_options {
      - enabled = false -> null
    }

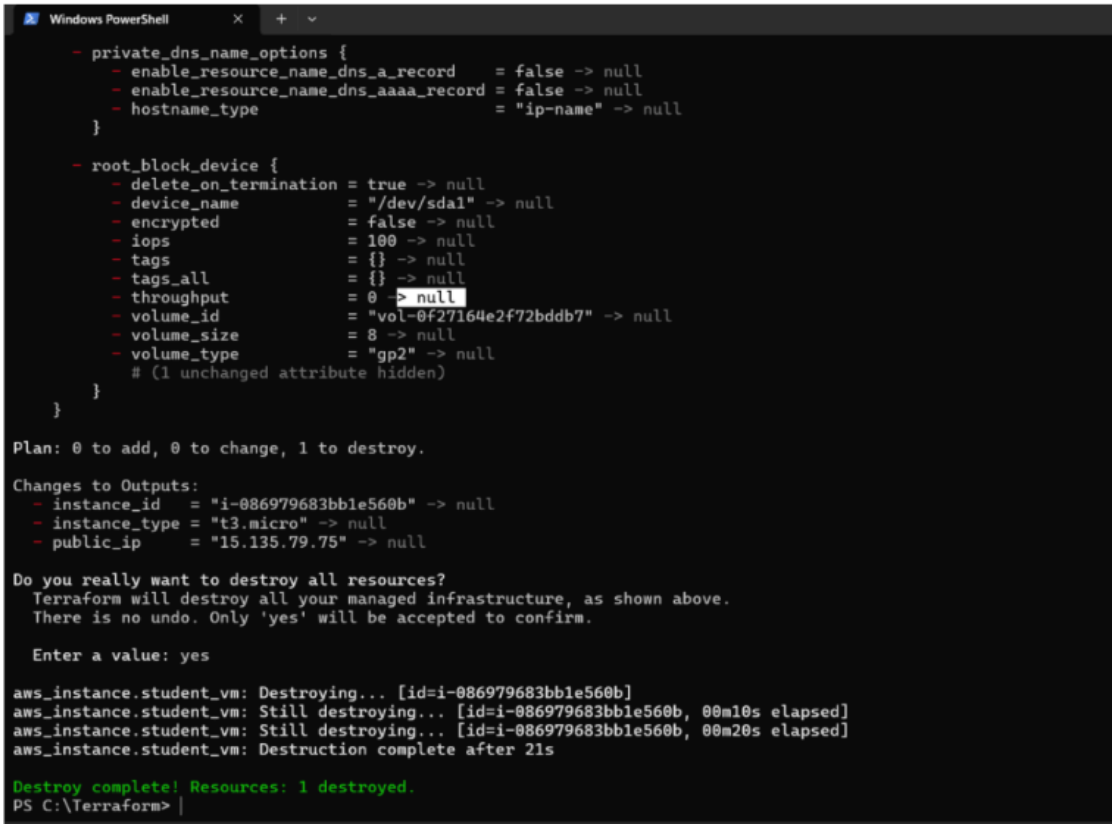
  - maintenance_options {
      - auto_recovery = "default" -> null
    }

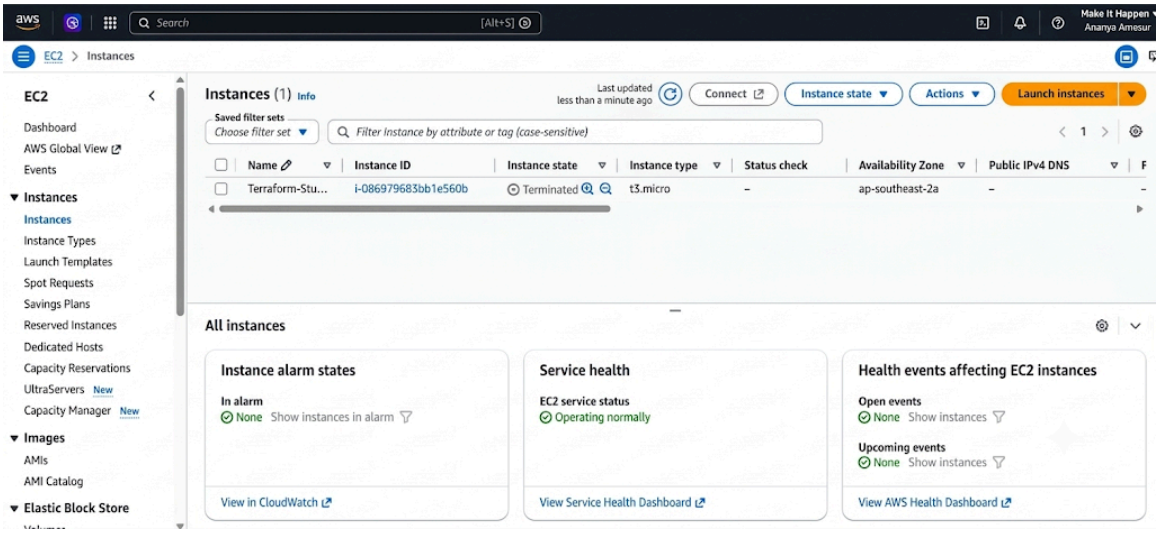
  - metadata_options {
      - http_endpoint               = "enabled" -> null
      - http_protocol_ipv6          = "disabled" -> null
      - http_put_response_hop_limit = 1 -> null
      - http_tokens                 = "optional" -> null
      - instance_metadata_tags      = "disabled" -> null
    }

  - primary_network_interface {
      - delete_on_termination = true -> null
      - network_interface_id  = "eni-012741dbd3bfef9ba" -> null
    }

  - private_dns_name_options {
      - enable_resource_name_dns_a_record    = false -> null
      - enable_resource_name_dns_aaaa_record = false -> null
      - hostname_type                        = "ip-name" -> null
    }

  - root_block_device {
      - delete_on_termination = true -> null
      - device_name           = "/dev/sda1" -> null
      - encrypted             = false -> null
      - iops                  = 100 -> null
      - tags                  = {} -> null
      - tags_all              = {} -> null
      - throughput            = 0 -> null
```





Infrastructure Attributes:

Parameter Name	Deployed Value	Source / Verification Method
AWS Account ID	702561770822	aws sts get-caller-identity

IAM User	Experiment7	AWS IAM Console / CLI
AWS Region	ap-southeast-2 (Sydney)	HCL Provider Configuration
Instance Name Tag	Terraform-Student-VM	main.tf HCL Block
Instance ID	i-086979683bb1e560b	terraform output / AWS Console
Instance Type	t3.micro	AWS EC2 Instance Details
Public IPv4 Address	15.135.79.75	terraform output / AWS Console
Private IPv4 Address	172.31.7.17	AWS EC2 Instance Details
VPC ID	vpc-0e56bf12ccd240a63	AWS VPC Console / data.aws_vpc
Subnet ID	subnet 0ecc646eb242b088a	AWS Subnet Console / data.aws_subnets
Key Pair	vmb-key	AWS EC2 Key Pairs / main.tf
Lifecycle Final State	Terminated	terraform destroy / AWS Console

Conclusion:

Cloud infrastructure was successfully provisioned, verified, and decommissioned on AWS using HashiCorp Terraform. A declarative configuration architecture was established utilizing HashiCorp Configuration Language (HCL), which dynamically resolved AWS Systems Manager (SSM) parameters, existing VPCs, and default subnets in the ap-southeast-2 region to deploy an EC2 instance. The complete Terraform execution lifecycle—spanning initialization (terraform init), syntax validation (terraform validate), state-aware planning (terraform plan), execution (terraform apply), and extraction of output parameters (terraform output)—was verified against live AWS Management Console resources. Finally, the automated teardown was performed using terraform destroy, demonstrating how Infrastructure as Code provides end-to-end automation, exact reproducibility, and efficient resource lifecycle control